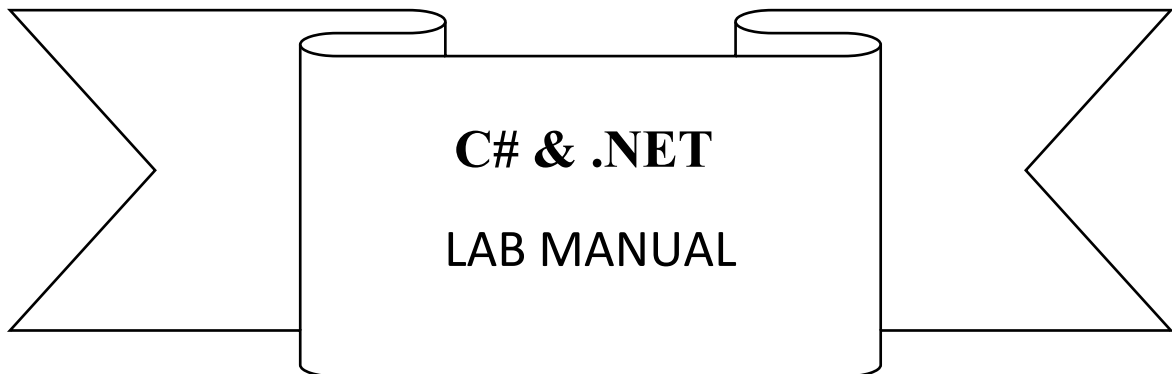


**NITTE MEENAKSHI INSTITUTE OF TECHNOLOGY**  
**DEPARTMENT OF MCA**



Academic Year  
2023-24 & 2024-25

| <b>Programming Using C# and .NET Laboratory</b><br>[As per Choice Based Credit System (CBCS) scheme]<br><b>SEMESTER – III</b>                                                                                                                                                                                                                                             |                   |           |    |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|-----------|----|
| Subject Code                                                                                                                                                                                                                                                                                                                                                              | 22MCA307L         | CIE Marks | 50 |
| Number of Hours/Week                                                                                                                                                                                                                                                                                                                                                      | 02 Hrs Laboratory | SEE Marks | 50 |
|                                                                                                                                                                                                                                                                                                                                                                           |                   | SEE Hours | 03 |
| <b>CREDITS – 01</b>                                                                                                                                                                                                                                                                                                                                                       |                   |           |    |
| <b>COURSE DESCRIPTION</b><br>To develop software development skills in C# programming and Students will have the proficiency to develop projects in C# programming. The course helps the students to solve the interdisciplinary applications through C# programming.<br>To write programs for solving real world problems using .NET Collection framework.               |                   |           |    |
| <b>PREREQUISITES</b> <ul style="list-style-type: none"> <li>• Basic understanding of Principles of Programming.</li> <li>• Basic understanding of Operating System, Networking, RDBMS and Files</li> </ul>                                                                                                                                                                |                   |           |    |
| <b>COURSE OBJECTIVES</b> <ul style="list-style-type: none"> <li>• Practice object-oriented programs and build .NET applications.</li> <li>• Implement C# programs for establishing interfaces.</li> <li>• Implement sample programs for developing reusable software components.</li> <li>• Create database connectivity in C# and implement GUI applications.</li> </ul> |                   |           |    |
| <b>Laboratory Experiments:</b>                                                                                                                                                                                                                                                                                                                                            |                   |           |    |
| <b>SECTION A</b>                                                                                                                                                                                                                                                                                                                                                          |                   |           |    |
| 1. Write a Program in C# to demonstrate Command line arguments processing for the following.                                                                                                                                                                                                                                                                              |                   |           |    |
| a) To find the square root of a given number.                                                                                                                                                                                                                                                                                                                             |                   |           |    |
| b) To find the sum & average of three numbers.                                                                                                                                                                                                                                                                                                                            |                   |           |    |
| 2. Write a Program in C# to demonstrate the following                                                                                                                                                                                                                                                                                                                     |                   |           |    |
| a) Boxing and Unboxing b) Invalid Unboxing.                                                                                                                                                                                                                                                                                                                               |                   |           |    |
| 3. Write a program in C# to add Two complex numbers using Operator overloading.                                                                                                                                                                                                                                                                                           |                   |           |    |
| 4. Write a Program in C# to find the sum of each row of given jagged array of 3 inner arrays.                                                                                                                                                                                                                                                                             |                   |           |    |
| 5. Write a Program in C# to demonstrate Array Out of Bound Exception using Try, Catch and Finally blocks.                                                                                                                                                                                                                                                                 |                   |           |    |
| 6. Write a Program to Demonstrate Use of Virtual and override key words in C# with a simple program.                                                                                                                                                                                                                                                                      |                   |           |    |
| 7. Write a Program in C# to create and implement a Delegate for any two arithmetic operations                                                                                                                                                                                                                                                                             |                   |           |    |
| 8. Write a Program in C# to demonstrate abstract class and abstract methods in C#.                                                                                                                                                                                                                                                                                        |                   |           |    |
| 9. Write a program to Set & Get the Name & Age of a person using Properties of C# to illustrate the use of different properties in C#.                                                                                                                                                                                                                                    |                   |           |    |
| 10. Write a Program in C# Demonstrate arrays of interface types (for runtime polymorphism).                                                                                                                                                                                                                                                                               |                   |           |    |
| 11. Consider the Database db_EMS (Employee Management System) consisting of the following tables :                                                                                                                                                                                                                                                                        |                   |           |    |
| tbl_Designations (IdDesignation: int, Designation: string)                                                                                                                                                                                                                                                                                                                |                   |           |    |
| tbl_EmployeeDetails (IdEmployee: int, Employee Name: string, Contact Number: string, Id Designation: int, IdReportingTo: int)                                                                                                                                                                                                                                             |                   |           |    |
| Develop a suitable window application using C#.NET having following options.                                                                                                                                                                                                                                                                                              |                   |           |    |

1. Enter new Employee details with designation & Reporting Manager.
2. Display all the Project Leaders (In a Grid) reporting to selected Project Managers (In a Combo box).
3. Display all the Engineers (In a Grid) reporting to selected Project Leader (In a Combo box).
4. Display all the Employees (In a Grid) with their reporting Manager (No Value for PM).

12. Consider the Database db\_LSA (Lecturer Subject Allocation) consisting of the following tables:

tbl\_Subjects(IdSubject: int, SubjectCode: string, SubjectName: string)

tbl\_Lecturers(IdLecturer: int, LecturerName: string, ContactNumber: string)

tbl\_LecturerSubjects(IdSubject: int, SubjectCode: string, IdLecturer: int)

Develop a suitable window application using C#.NET having following options.

1. Enter new Subject Details.
2. Enter New Lecturer Details.
3. Subject Allocation with Lecturer Name in a Combo box and subjects to be allocated in Grid with checkbox Column.
4. Display all the subjects allocated (In a Grid) to the selected Lecturer (In a Combo Box).

13. Consider the database db\_VSS (Vehicle Service Station) consisting of the following tables:

tbl\_VehicleTypes(IdVehicleType: int, VehicleType: string, ServiceCharge: int)

tbl\_ServiceDetails(IdService: int, VehicleNumber: string, ServiceDetails: string, IdVehicleType: int)

Develop a suitable window application using C#.NET having following options.

1. Enter new Service Details for the Selected Vehicle Type (In a Combo Box).
2. Update the Existing Service Charges to Database.
3. Total Service Charges Collected for the Selected Vehicle (In a Combo box) with total amount displayed in a text box.

### TEACHING METHODS

- PPTs
- Hands-On Sessions Based Teaching
- Visual Studio 2008 and above

### ASSESSMENT METHODS

|                                        |    |
|----------------------------------------|----|
| Experiment Write up + Execution + Viva | 20 |
| Lab Record Writing                     | 15 |
| Lab Internals Test                     | 15 |
| Total                                  | 50 |

Final Exam will be conducted for 50 marks (SEE).

**Course Outcome (CO): At the end of this course, the students will be able to:**

**CO1:** Understand C# and client-server concepts using .Net Frame Work Components

**CO2:** Apply delegates, event and exception handling to incorporate with ASP, Win Form, ADO.NET

**CO3:** Analyze the use of .Net Components depending on the problem statement

**CO4:** Implement & develop a web based and Console based application with Database connectivity

### Instructions to execute command line argument program

- 1) Write the above program.

- 2) Goto **Debug Menu->ConsoleApp1 (this is project name) Debug Properties.**
- 3) **Launch profile** window will be opened -> under **command line arguments** enter the value and close the window.
- 4) Then execute the program using **run button** like other program.
- 5) If the input contains multiple arguments then use **Enter key** or **Space bar** to enter values.

### **1a) Square root**

```
namespace Cons
{
    public class squareroot
    {
        public static void Main(String[] args)
        {
            int Number = Convert.ToInt16(args[0]);
            double SqrtNumber = Math.Sqrt(Number);
            Console.WriteLine("Square root of {0} is: {1}", Number, SqrtNumber);
            Console.ReadLine();
        }
    }
}
```

### **1b) Find the Sum and Average**

```
namespace lab1b
{
    class Program
    {
        static void Main(string[] args)
        {
            int a, b, c, sum;
            float avg;
            a = int.Parse(args[0]);
            b = int.Parse(args[1]);
            c = int.Parse(args[2]);
            sum = a + b + c;
            Console.WriteLine("The sum is:{0} ", sum);
            Console.ReadLine();
            avg = sum / 3;
            Console.WriteLine("The avg of given sum is:{0} ", avg);
            Console.ReadLine();
        }
    }
}
```

2. Write a Program in C# to demonstrate the following :
  - a) Boxing and Unboxing

b) Invalid Unboxing.

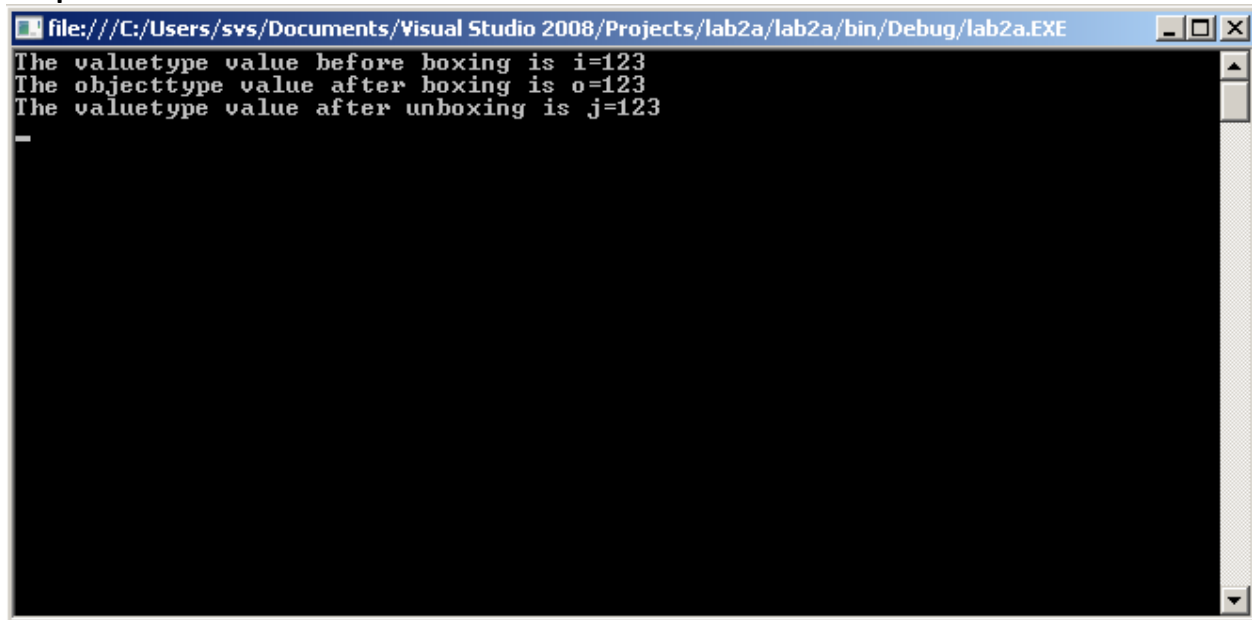
### **Solution:**

#### **a) Boxing and Unboxing**

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Text;
using System.Threading.Tasks;

namespace lab2a
{
    class Program
    {
        static void Main(string[] args)
        {
            int i = 123;
            object o = i;
            int j = (int)o;
            Console.WriteLine("The valuetype value before boxing is i=" + i);
            Console.WriteLine("The objecttype value after boxing is o=" + o);
            Console.WriteLine("The valuetype value after unboxing is j=" +
j);
            Console.ReadLine();
        }
    }
}
```

#### **Output:**

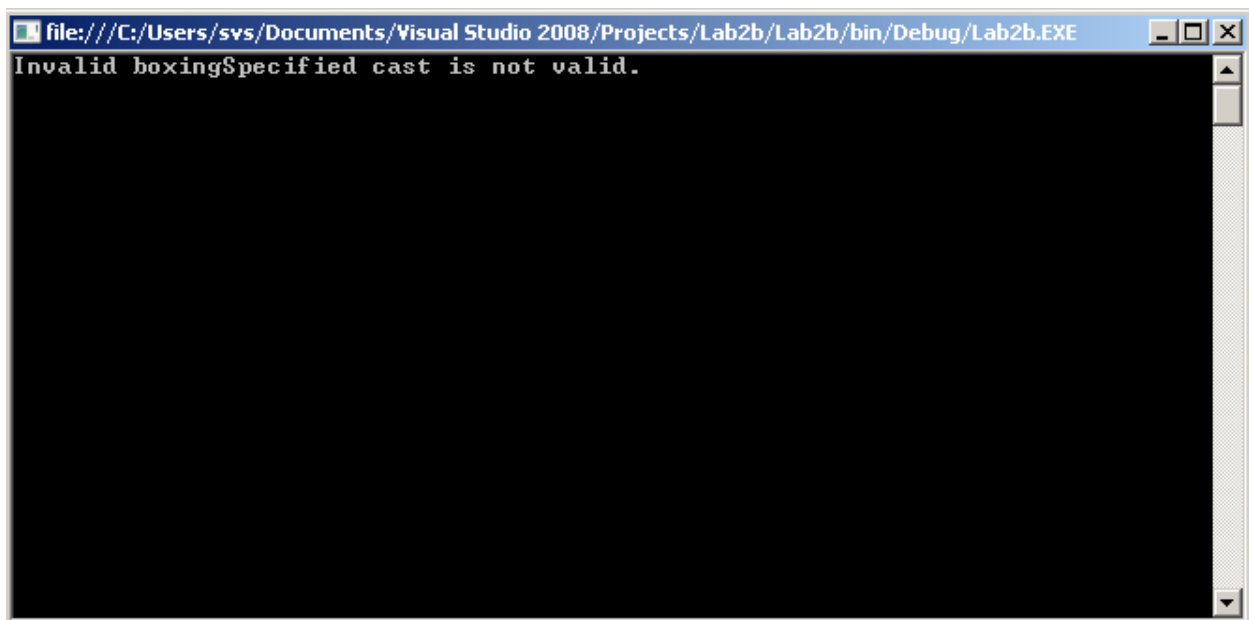
A screenshot of a Windows console window titled "file:///C:/Users/svs/Documents/Visual Studio 2008/Projects/lab2a/lab2a/bin/Debug/lab2a.EXE". The console displays three lines of text: "The valuetype value before boxing is i=123", "The objecttype value after boxing is o=123", and "The valuetype value after unboxing is j=123". The text is in a monospaced font, and the background is black with white text. The window has standard Windows window controls (minimize, maximize, close) in the top right corner.

b) Invalid Unboxing.

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Text;

namespace lab2b
{
    class Program
    {
        static void Main(string[] args)
        {
            int i = 123;
            object o = i;
            try
            {
                int j = (Short)o;
                Console.WriteLine("unboxing is successful");
            }
            catch (System.InvalidCastException e)
            {
                Console.WriteLine("Invalid boxing" + e.Message);
            }
            Console.ReadLine();
        }
    }
}
```

**Output:**



**Here, if we change the data type of o to int i.e**

```
int j = (int)o;
```

Then we can achieve unboxing.

Output:



The image shows a screenshot of a Windows command prompt window. The title bar at the top reads "file:///C:/Users/svs/Documents/Visual Studio 2008/Projects/Lab2b/Lab2b/bin/Debug/Lab2b.EXE". The main area of the window is black, and the text "unboxing is successful" is displayed in white at the top left. A small white cursor is visible on the line below the text. The window has standard Windows window controls (minimize, maximize, close) in the top right corner.

3. Write a program in C# to add Two complex numbers using Operator overloading.

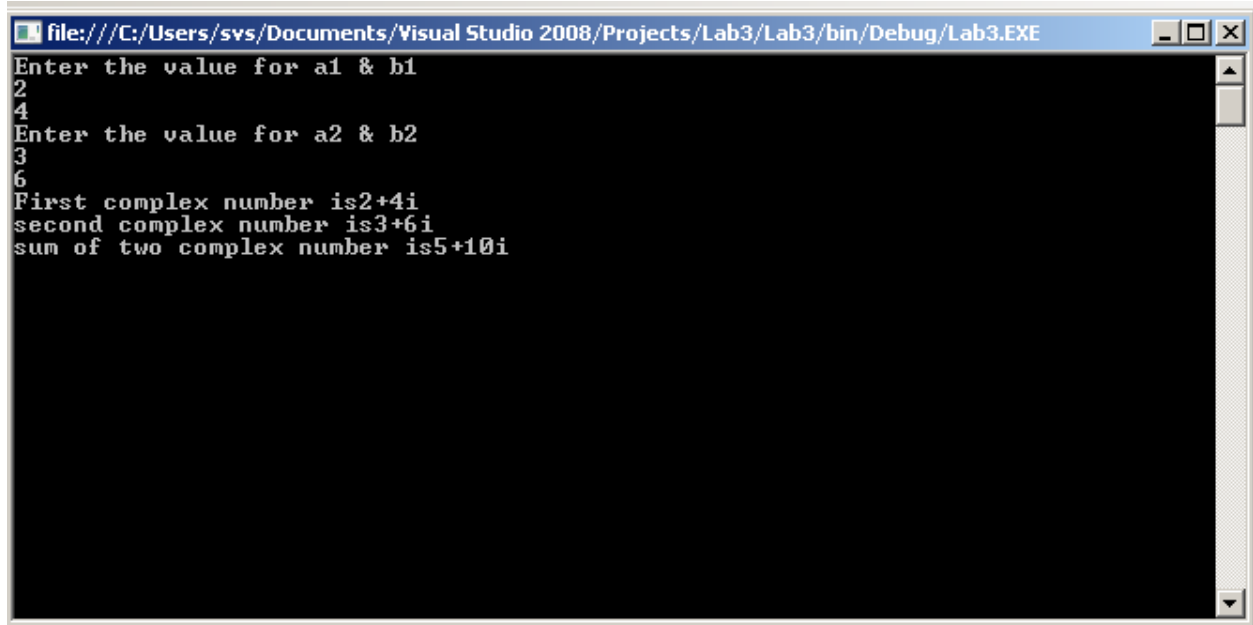
**Solution:****To add two complex numbers using Operator overloading:**

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Text;

namespace lab3
{
    class complex
    {
        int real;
        int imaginary;
        public complex(int r, int img)
        {
            real = r;
            imaginary = img;
        }
        public static complex operator +(complex c1, complex c2)
        {
            return new complex(c1.real + c2.real, c1.imaginary +
c2.imaginary);
        }
        public override string ToString()
        {
            return (String.Format("{0}+{1}i", real, imaginary));
        }
    }
    class demo
    {
        static void Main(string[] args)
        {
            int a1, b1, a2, b2;
            Console.WriteLine("Enter the value for a1 & b1");
            a1 = Int32.Parse(Console.ReadLine());
            b1 = Int32.Parse(Console.ReadLine());
            complex num1 = new complex(a1, b1);
            Console.WriteLine("Enter the value for a2 & b2");
            a2 = Int32.Parse(Console.ReadLine());
            b2 = Int32.Parse(Console.ReadLine());
            complex num2 = new complex(a2, b2);
            complex sum = num1 + num2;
            Console.WriteLine("First complex number is" + num1);
            Console.WriteLine("second complex number is" + num2);
            Console.WriteLine("sum of two complex number is" + sum);
            Console.ReadLine();
        }
    }
}
```

**Output:**





```
file:///C:/Users/svs/Documents/Visual Studio 2008/Projects/Lab3/Lab3/bin/Debug/Lab3.EXE
Enter the value for a1 & b1
2
4
Enter the value for a2 & b2
3
6
First complex number is2+4i
second complex number is3+6i
sum of two complex number is5+10i
```

4. Write a Program in C# to find the sum of each row of given jagged array of 3 inner arrays.

## **Solution:**

**To find the sum of each row of given jagged array of 3 inner arrays:**

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Text;

namespace lab4
{
    class jaggedarrays
    {
        int[][] jagged = new int[3][];
        public void ReadArrays()
        {
            Console.Write("Enter the size of first inner array:");
            jagged[0] = new int[int.Parse(Console.ReadLine())];
            Console.WriteLine("enter the elements of first inner array:");
            for (int i = 0; i < jagged[0].Length; i++)
                jagged[0][i] = int.Parse(Console.ReadLine());
            Console.Write("Enter the size second of inner array:");
            jagged[1] = new int[int.Parse(Console.ReadLine())];
            Console.WriteLine("enter the elements of second inner array:");
            for (int i = 0; i < jagged[1].Length; i++)
                jagged[1][i] = int.Parse(Console.ReadLine());
            Console.Write("Enter the size third of inner array:");
            jagged[2] = new int[int.Parse(Console.ReadLine())];
            Console.WriteLine("enter the elements of third inner array:");
            for (int i = 0; i < jagged[2].Length; i++)
                jagged[2][i] = int.Parse(Console.ReadLine());
        }
        public void FindSum()
        {
            int sum = 0;
            for (int i = 0; i < jagged[0].Length; i++)
                sum = sum + jagged[0][i];
            Console.WriteLine("\n\n Sum of all the first inner array
is={0}", sum);

            sum = 0;
            for (int i = 0; i < jagged[1].Length; i++)
                sum = sum + jagged[1][i];
            Console.WriteLine("\n\n Sum of all the second inner array
is={0}", sum);

            sum = 0;
            for (int i = 0; i < jagged[2].Length; i++)
                sum = sum + jagged[2][i];
            Console.WriteLine("\n\n Sum of all the third inner array
is={0}", sum);
        }
        public void PrintArrays()
    }
}
```

```

    {
        Console.WriteLine("\nElements of first inner array:");
        for (int i = 0; i < jagged[0].Length; i++)
            Console.WriteLine(jagged[0][i] + "\t");
        Console.WriteLine("\nElements of second inner array:");
        for (int i = 0; i < jagged[1].Length; i++)
            Console.WriteLine(jagged[1][i] + "\t");
        Console.WriteLine("\nElements of third inner array:");
        for (int i = 0; i < jagged[2].Length; i++)
            Console.WriteLine(jagged[2][i] + "\t");
    }
}
class demo
{
    static void Main(string[] args)
    {
        jaggedarrays ja = new jaggedarrays();
        ja.ReadArrays();
        ja.PrintArrays();
        ja.FindSum();
        Console.ReadLine();
    }
}
}

```

#### Output:

```

file:///C:/Users/svs/Documents/Visual Studio 2008/Projects/Lab4/Lab4/bin/Debug/Lab4.EXE
1
2
3
Enter the size third of inner array:4
enter the elements of third inner array:
5
6
7
8

Elements of first inner array:3 2
Elements of second inner array:1      2      3
Elements of third inner array:5 6      7      8

Sum of all the first inner array is=5

Sum of all the second inner array is=6

Sum of all the third inner array is=26

```

5. Write a Program in C# to demonstrate Array Out of Bound Exception using Try, Catch and Finally blocks.

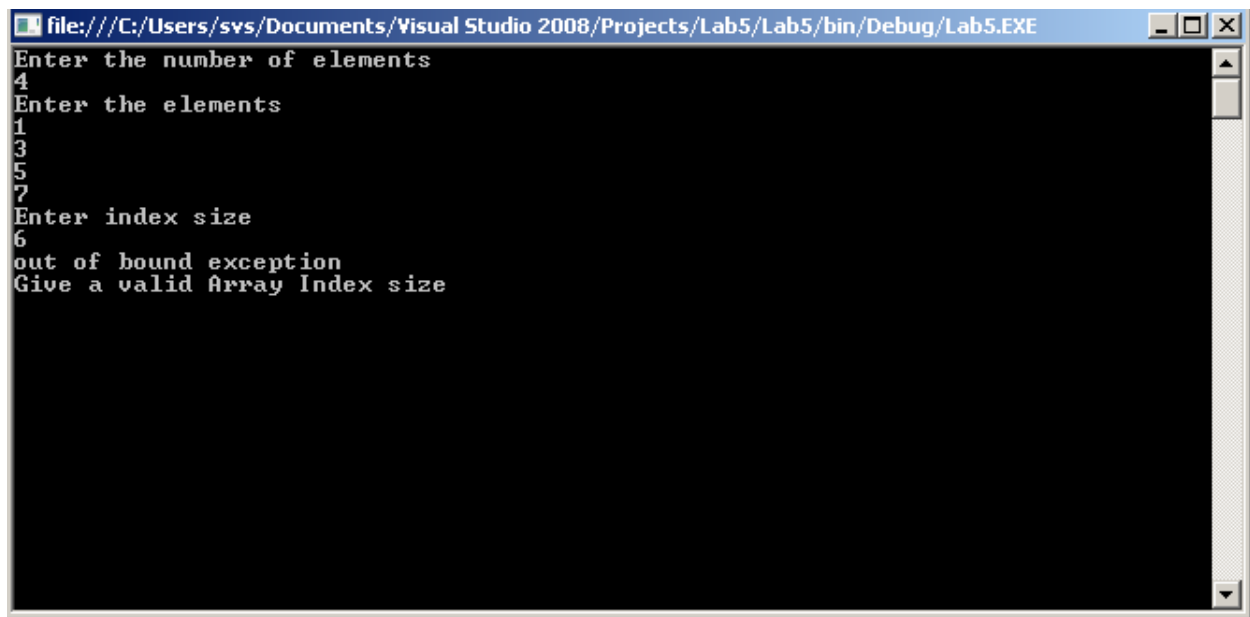
#### Solution:

## To demonstrate Array Out of Bound Exception using Try, Catch and Finally blocks:

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Text;

namespace lab5
{
    class Program
    {
        public static void Main(string[] args)
        {
            int i, ele;
            Console.WriteLine("Enter the number of elements");
            ele = int.Parse(Console.ReadLine());
            int[] a = new int[ele];
            Console.WriteLine("Enter the elements");
            for (i = 0; i != ele; i++)
            {
                a[i] = int.Parse(Console.ReadLine());
            }
            Console.WriteLine("Enter index size");
            int element = int.Parse(Console.ReadLine());
            try
            {
                Console.WriteLine("element is" + a[element]);
            }
            catch (IndexOutOfRangeException e)
            {
                Console.WriteLine("out of bound exception");
            }
            finally
            {
                Console.WriteLine("Give a valid Array Index size");
            }
            Console.ReadLine();
        }
    }
}
```

## Output:



```
file:///C:/Users/svs/Documents/Visual Studio 2008/Projects/Lab5/Lab5/bin/Debug/Lab5.EXE
Enter the number of elements
4
Enter the elements
1
3
5
7
Enter index size
6
out of bound exception
Give a valid Array Index size
```

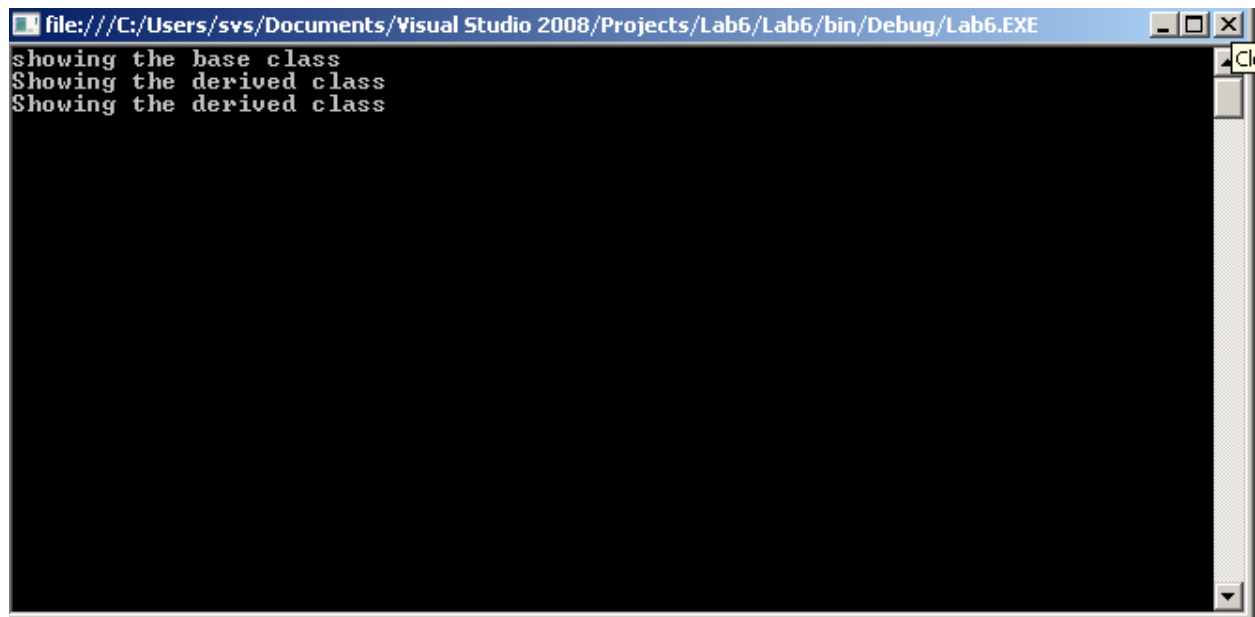
6. Write a Program to Demonstrate Use of Virtual and override key words in C# with a simple program.

**Solution:****To Demonstrate Use of Virtual and override key words in C# with a simple program:**

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Text;

namespace lab6
{
    class Base
    {
        public virtual void show()
        {
            Console.WriteLine("showing the base class");
        }
    }
    class Derived : Base
    {
        public override void show()
        {
            Console.WriteLine("Showing the derived class");
        }
    }
    class demo
    {
        static void Main(string[] args)
        {
            Base b = new Base();
            b.show();
            Derived d = new Derived();
            d.show();
            Base b1 = new Derived();
            b1.show();
            Console.ReadLine();
        }
    }
}
```

**Output:**



```
file:///C:/Users/svs/Documents/Visual Studio 2008/Projects/Lab6/Lab6/bin/Debug/Lab6.EXE
showing the base class
Showing the derived class
Showing the derived class
```

7. Write a Program in C# to create and implement a Delegate for any two arithmetic operations

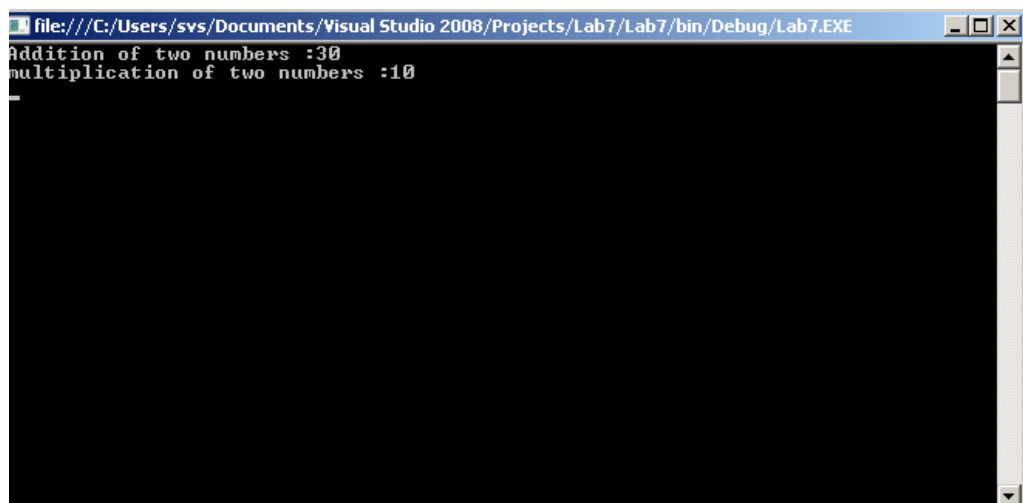
### **Solution:**

#### **To create and implement a Delegate for any two arithmetic operations:**

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Text;

public delegate int operation(int p, int q);
namespace lab7
{
    class delegates
    {
        public static int AddNum(int p, int q)
        {
            return (p + q);
        }
        public int MultiNum(int p, int q)
        {
            return (p * q);
        }
        static void Main(string[] args)
        {
            operation op1 = new operation(AddNum);
            Console.WriteLine("Addition of two numbers :{0}", op1(10, 20));
            delegates d = new delegates();
            operation op2 = new operation(d.MultiNum);
            Console.WriteLine("multiplication of two numbers :{0}", op2(5,2));
            Console.ReadLine();
        }
    }
}
```

### **Output:**



```
file:///C:/Users/svs/Documents/Visual Studio 2008/Projects/Lab7/Lab7/bin/Debug/Lab7.EXE
Addition of two numbers :30
multiplication of two numbers :10
```

8. Write a Program in C# to demonstrate abstract class and abstract methods in C#.

### **Solution**

#### **To demonstrate abstract class and abstract methods in C#:**

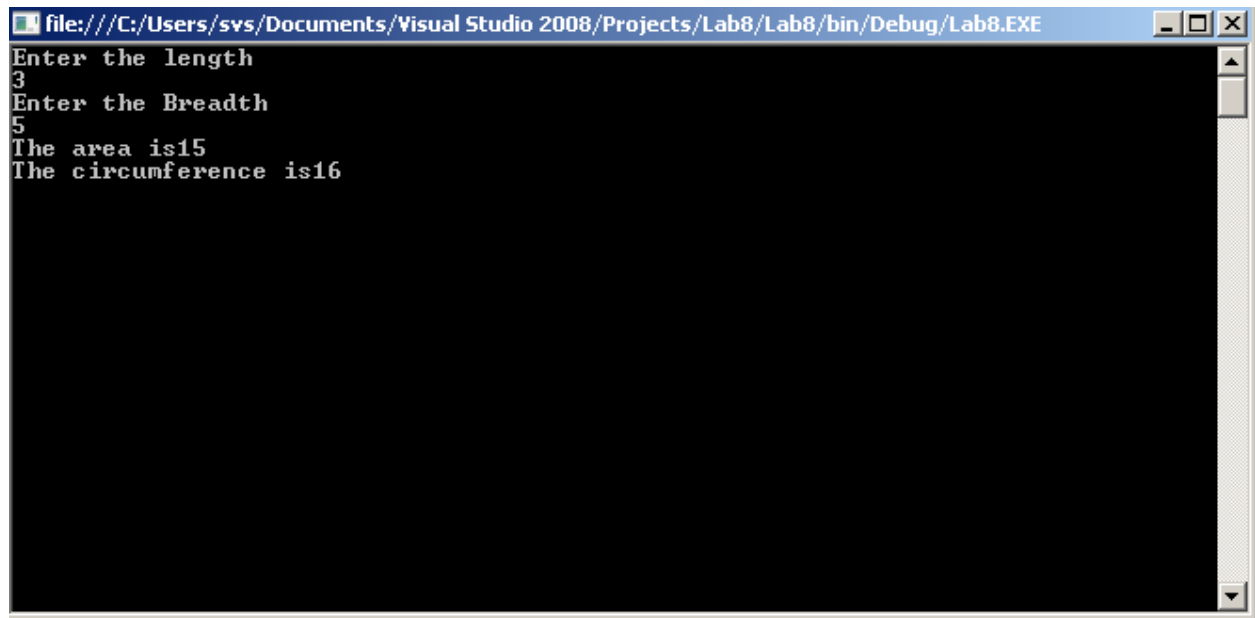


```

using System;
using System.Collections.Generic;
using System.Linq;
using System.Text;
namespace lab8
{
    abstract class shape
    {
        protected float l, b;
        public abstract float Area();
        public abstract float Circumference();
    }
    class Rectangle : shape
    {
        public void getlb()
        {
            Console.WriteLine("Enter the length");
            l = float.Parse(Console.ReadLine());
            Console.WriteLine("Enter the Breadth");
            b = float.Parse(Console.ReadLine());
        }
        public override float Area()
        {
            return l * b;
        }
        public override float Circumference()
        {
            return 2 * (l + b);
        }
    }
    class demo
    {
        static void Main(string[] args)
        {
            Rectangle r = new Rectangle();
            r.getlb();
            Console.WriteLine("The area is" + r.Area());
            Console.WriteLine("The circumference is" + r.Circumference());
            Console.ReadLine();
        }
    }
}

```

**Output:**



The screenshot shows a Windows command prompt window with a blue title bar. The title bar text is "file:///C:/Users/svs/Documents/Visual Studio 2008/Projects/Lab8/Lab8/bin/Debug/Lab8.EXE". The command prompt has a black background with white text. The text displayed is: "Enter the length", "3", "Enter the Breadth", "5", "The area is15", and "The circumference is16". The input "3" is on the same line as "Enter the length", and "5" is on the same line as "Enter the Breadth".

```
file:///C:/Users/svs/Documents/Visual Studio 2008/Projects/Lab8/Lab8/bin/Debug/Lab8.EXE
Enter the length
3
Enter the Breadth
5
The area is15
The circumference is16
```

9. Write a program to Set & Get the Name & Age of a person using Properties of C# to illustrate the use of different properties in C#.

**Solution:**

## To Set & Get the Name & Age of a person using Properties of C# to illustrate the use of different properties in C#:

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Text;

namespace lab9
{
    class Person
    {
        string name;
        int age;
        public string pername
        {
            get { return name; }
            set { name = value; }
        }
        public int perage
        {
            get { return age; }
            set { age = value; }
        }
    }
    class demo
    {
        static void Main(string[] args)
        {
            Person p1 = new Person();
            Console.WriteLine("Enter the name of a person");
            p1.pername = Console.ReadLine();
            Console.WriteLine("Enter the age of a person");
            p1.perage = Convert.ToInt32(Console.ReadLine());
            Console.WriteLine("Name of a person is:" + p1.pername);
            Console.WriteLine("Age of a person is:" + p1.perage);
            Console.ReadLine();
        }
    }
}
```

## Output:



A screenshot of a Windows command prompt window. The title bar at the top reads "file:///C:/Users/svs/Documents/Visual Studio 2008/Projects/Lab9/Lab9/bin/Debug/Lab9.EXE". The window has standard Windows window controls (minimize, maximize, close) on the right. The command prompt area is black with white text. The text displayed is as follows:

```
Enter the name of a person
Ashok
Enter the age of a person
22
Name of a person is:Ashok
Age of a person is:22
```

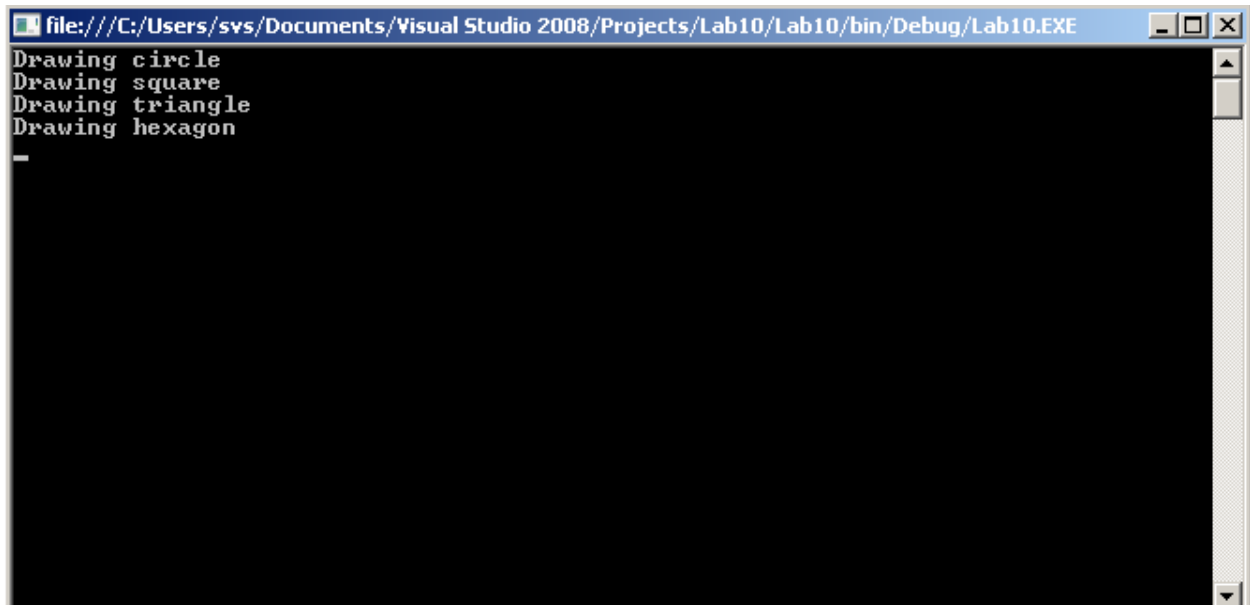
10. Write a Program in C# Demonstrate arrays of interface types (for runtime polymorphism).

**Solution:****To demonstrate arrays of interface types (for runtime polymorphism):**

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Text;

namespace lab10
{
    interface ishape
    {
        void draw();
    }
    class circle : ishape
    {
        public void draw()
        {
            Console.WriteLine("Drawing circle");
        }
    }
    class square : ishape
    {
        public void draw()
        {
            Console.WriteLine("Drawing square");
        }
    }
    class triangle : ishape
    {
        public void draw()
        {
            Console.WriteLine("Drawing triangle");
        }
    }
    class hexagon : ishape
    {
        public void draw()
        {
            Console.WriteLine("Drawing hexagon");
        }
    }
    class Program
    {
        static void Main(string[] args)
        {
            ishape[] shape = { new circle(), new square(), new triangle(),
new hexagon() };
            foreach (ishape s in shape)
                s.draw();
            Console.ReadLine();
        }
    }
}
```

**Output:**



A screenshot of a Windows command prompt window. The title bar at the top reads "file:///C:/Users/svs/Documents/Visual Studio 2008/Projects/Lab10/Lab10/bin/Debug/Lab10.EXE". The window has standard Windows window controls (minimize, maximize, close) on the right. The main area is black with white text. The text displayed is:

```
Drawing circle  
Drawing square  
Drawing triangle  
Drawing hexagon  
-
```

The cursor is positioned at the end of the last line, after the hyphen.

### Program:

1. Consider the Database db\_EMS (Employee Management System) consisting of the following tables :

tbl\_Designations (IdDesignation: int, Designation: string)

tbl\_EmployeeDetails (IdEmployee: int, EmployeeName: string, ContactNumber: string, IdDesignation: int, IdReportingTo: int)

Develop a suitable window application using C#.NET having following options.

1. Enter new Employee details with designation & Reporting Manager.
2. Display all the Project Leaders (In a Grid) reporting to selected Project Managers (In a Combo box).
3. Display all the Engineers (In a Grid) reporting to selected Project Leader (In a Combo box).
4. Display all the Employees (In a Grid) with their reporting Manager (No Value for PM).

NOTE: tbl\_Designation is a static table containing the following Rows in it.

- 1 Project Manager
- 2 Project Leader
- 3 Engineer

### Form Design

|   | IdEmployee | EmployeeName | ContactNumber | IdDesignation | IdReportingTo |
|---|------------|--------------|---------------|---------------|---------------|
| ▶ | 100        | nithin rai b | 7878787       | 1             | 3             |
|   | 2          | Ashish       | 2345          | 2             | 3             |
|   | 45         | Advit        | 890           | 1             | 1             |
|   | 6          | Suvit        | 8976          | 1             | 2             |

### C# Code:

```
using System.Data;  
using System.Data.SqlClient;  
using System.Collections.Generic;  
using System.ComponentModel;  
using System.Drawing;
```

```

using System.Linq;
using System.Text;
using System.Windows.Forms;
using static System.Windows.Forms.VisualStyles.VisualStyleElement;
using System.Diagnostics;

namespace WinFormsApp2
{
    public partial class Form1 : Form
    {
        SqlConnection con = new SqlConnection(@"Data Source=NITHIN-LAP\SQLEXPRESS;Initial Catalog = Student; Integrated Security = True");
        SqlCommand cmd = new SqlCommand();
        SqlDataReader dr;
        SqlDataAdapter da;
        DataSet ds = new DataSet();
        DataTable dt = new DataTable();
        public Form1()
        {
            InitializeComponent();
        }

        public void loadlist()
        {
            con.Open();
            cmd.CommandText = "select * from tbl_Designations";
            dr = cmd.ExecuteReader();
            if (dr.HasRows)
            {
                while (dr.Read())
                {
                    comboBox1.Items.Add(dr[0].ToString());
                    comboBox2.Items.Add(dr[1].ToString());
                }
            }
            con.Close();
        }

        private void button1_Click(object sender, EventArgs e)
        {
            con.Open();

            cmd.Connection = con;

            cmd.CommandText = "insert into tbl_EmployeeDetails(IdEmployee, EmployeeName, ContactNumber, IdDesignation, IdReportingTo) values('" +
            textBox1.Text + "','" + textBox2.Text + "','" + textBox3.Text + "','" +
            comboBox1.Text + "','" + textBox4.Text + "')";
            cmd.ExecuteNonQuery();
            MessageBox.Show("Record inserted");
            con.Close();
            textBox1.Text = "";
            textBox2.Text = "";
            textBox3.Text = "";
            textBox4.Text = "";
        }

        private void comboBox2_SelectedIndexChanged(object sender, EventArgs e)
        {

```



```

        if (comboBox2.SelectedIndex == 0)
        {
            dt.Reset();
            con.Open();
            ds.Reset();
            ds.Tables.Add(dt);
            da = new SqlDataAdapter("select * from tbl_EmployeeDetails ", con);
            da.Fill(dt);
            dataGridView1.DataSource = dt.DefaultView;
            con.Close();
        }
        else
        {
            con.Open();
            ds.Reset();
            ds.Tables.Add(dt);
            da = new SqlDataAdapter("select * from tbl_EmployeeDetails where IdDesignation = '" +
comboBox2.SelectedIndex + "'", con);

            da.Fill(dt);
            dataGridView1.DataSource = dt.DefaultView;
            con.Close();
        }
    }

    private void Form1_Load_1(object sender, EventArgs e)
    {
        cmd.Connection = con;
        loadlist();
    }
}

```

12. Consider the Database db\_LSA (Lecturer Subject Allocation) consisting of the following tables:

tbl\_Subjects(IdSubject: int, SubjectName: string)

tbl\_Lecturers(IdLecturer: int, LecturerName: string)

tbl\_LecturerSubjects(LecturerName: string, SubjectName: string)

Develop a suitable window application using C#.NET having following options.

1. Enter new Subject Details.
2. Enter New Lecturer Details.
3. Subject Allocation with Lecturer Name in a Combo box and subjects to be allocated in Grid with checkbox Column.
4. Display all the subjects allocated (In a Grid) to the selected Lecturer (In a Combo Box).

### Form Design

The screenshot displays a Windows Form titled "Form1" with a light gray background and standard Windows window controls (minimize, maximize, close) in the top right corner. The form is organized into four main sections arranged in a 2x2 grid:

- Subject Details:** Located in the top-left quadrant, it contains two text input fields labeled "Subject\_ID" and "Subject\_Name", and a gray "Insert" button below them.
- Lecturer Details:** Located in the top-right quadrant, it contains two text input fields labeled "Lecture\_ID" and "Lecture\_Name", and a gray "Insert" button below them.
- Subject Allocation:** Located in the bottom-left quadrant, it features a "Lecture\_Name" dropdown menu and a "Subject\_Name" label next to a large gray rectangular area representing a grid.
- Allocated Lectures:** Located in the bottom-right quadrant, it features a "Lecture\_Name" dropdown menu and a large gray rectangular area representing a grid.

At the bottom center of the form, there is a gray "Save" button.

### OUTPUT

Form1

Subject Details

Subject\_ID

6

Subject\_Name

DS

Insert

Lecturer Details

Lecture\_ID

Lecture\_Name

Insert

Subject Allocation

Lecture\_Name

Subject\_Name

|  | cbc                      | SubjectName |
|--|--------------------------|-------------|
|  | <input type="checkbox"/> | Java        |
|  | <input type="checkbox"/> | C#          |
|  | <input type="checkbox"/> | C++         |

Save

Lectures

Lecture\_ID

Lecture\_Name

Record inserted

OK

Form1

Subject Details

Subject\_ID

Subject\_Name

Insert

Lecturer Details

Lecture\_ID

13

Lecture\_Name

ABC

Insert

Subject Allocation

Lecture\_Name

Subject\_Name

|  | cbc                      | SubjectName |
|--|--------------------------|-------------|
|  | <input type="checkbox"/> | Java        |
|  | <input type="checkbox"/> | C#          |
|  | <input type="checkbox"/> | C++         |

Save

Lectures

Lecture\_ID

Lecture\_Name

Record inserted

OK

**Form1**

**Subject Details**

Subject\_ID

Subject\_Name

**Insert**

**Subject Allocation**

Lecture\_Name **Richard**

Subject\_Name

|   | cbc                                 | SubjectName |
|---|-------------------------------------|-------------|
|   | <input type="checkbox"/>            | Java        |
|   | <input type="checkbox"/>            | C#          |
| ▶ | <input checked="" type="checkbox"/> | C++         |
|   | <input type="checkbox"/>            |             |

**Save**

**Lecturer Details**

Lecture\_ID

Lecture\_Name

**Insert**

**Lectures**

inserted successfully

**OK**

**Form1**

**Subject Details**

Subject\_ID

Subject\_Name

**Insert**

**Subject Allocation**

Lecture\_Name **Richard**

Subject\_Name

|   | cbc                                 | SubjectName |
|---|-------------------------------------|-------------|
|   | <input type="checkbox"/>            | Java        |
|   | <input type="checkbox"/>            | C#          |
| ▶ | <input checked="" type="checkbox"/> | C++         |
|   | <input type="checkbox"/>            |             |

**Save**

**Lecturer Details**

Lecture\_ID

Lecture\_Name

**Insert**

**Allocated Lectures**

Lecture\_Name **Sowmya**

|   | LecturerName | SubjectName |
|---|--------------|-------------|
| ▶ | Sowmya       | C#          |
|   | Sowmya       | C++         |
| * |              |             |

### C# Code

```

using System.Data;
using System.Data.SqlClient;

namespace WinFormsApp3
{
    public partial class Form1 : Form
    {
        SqlConnection con = new SqlConnection(@"Data Source=NITHIN-LAP\SQLEXPRESS;Initial Catalog =
master; Integrated Security = True");
        SqlCommand cmd = new SqlCommand();
        SqlDataReader dr;
        SqlDataAdapter da;
    }
}

```

```

DataSet ds = new DataSet();
DataTable dt = new DataTable();

public Form1()
{
    InitializeComponent();
}

private void Form1_Load(object sender, EventArgs e)
{
    dt.Reset();
    con.Open();
    ds.Reset();
    ds.Tables.Add(dt);
    da = new SqlDataAdapter("select SubjectName from tbl_Subjects", con);
    da.Fill(dt);
    dataGridView1.DataSource = dt.DefaultView;
    con.Close();
    cmd.Connection = con;
    DataGridViewCheckBoxColumn cb = new DataGridViewCheckBoxColumn();
    cb.Name = "cbc";
    dataGridView1.Columns.Insert(0, cb);
    loadlist();
}

public void loadlist()
{
    con.Open();
    cmd.CommandText = "select * from tbl_Lecturers";
    dr = cmd.ExecuteReader();
    if (dr.HasRows)
    {
        while (dr.Read())
        {
            comboBox1.Items.Add(dr[1].ToString());
            comboBox2.Items.Add(dr[1].ToString());
        }
    }
    con.Close();
}

private void button2_Click(object sender, EventArgs e)
{
    con.Open();
    cmd.CommandText = "insert into tbl_Lecturers values(" + textBox3.Text
+ "," + textBox4.Text + ")";
    cmd.ExecuteNonQuery();
    cmd.Clone();
    MessageBox.Show("Record inserted");
    con.Close();
    textBox3.Text = "";
    textBox4.Text = "";
}

private void button1_Click(object sender, EventArgs e)
{

```

```

cmd.Connection = con;
con.Open();
cmd.CommandText = "insert into tbl_Subjects values('" + textBox1.Text
+ "','" + textBox2.Text + "')";
cmd.ExecuteNonQuery();
cmd.Clone();
MessageBox.Show("Record inserted");
con.Close();
textBox1.Text = "";
textBox2.Text = "";
}

private void comboBox2_SelectedIndexChanged(object sender, EventArgs e)
{
    DataSet ds1 = new DataSet();
    DataTable dt1 = new DataTable();
    con.Open();
    ds1.Reset();
    ds1.Tables.Add(dt1);
    SqlDataAdapter da1 = new SqlDataAdapter("select * from tbl_LecturerSubjects where LecturerName =
'" + comboBox2.SelectedItem.ToString() + "'", con);

    da1.Fill(dt1);
    dataGridView2.DataSource = dt1.DefaultView;
    con.Close();
}

private void button3_Click(object sender, EventArgs e)
{
    int i = 0;

    foreach (DataGridViewRow row in dataGridView1.Rows)
    {
        bool iss = Convert.ToBoolean(row.Cells["cbc"].Value);
        if (iss)
        {
            cmd.CommandText = "insert into tbl_LecturerSubjects values('" +
comboBox1.SelectedItem.ToString() + "','" + row.Cells[1].Value + "')";

            con.Open();
            cmd.ExecuteNonQuery();
            con.Close();
            MessageBox.Show("inserted successfully");
        }
    }
    i++;
}
}
}

```

**13) Consider the database db\_VSS (Vehicle Service Station) consisting of the following tables:**

**tbl\_VehicleTypes(IdVehicleType: int, VehicleType: string, ServiceCharge: int)**  
**tbl\_ServiceDetails(IdService: int, VehicleNumber: string, ServiceDetails: string, IdVehicleType: int)**

Develop a suitable window application using C#.NET having following options.

1. Enter new Service Details for the Selected Vehicle Type (In a Combo Box).
2. Update the Existing Service Charges to Database.
3. Total Service Charges Collected for the Selected Vehicle (In a Combo box) with total amount displayed in a text box. NOTE: tbl\_VehicleType is a static table containing the following Rows in it. 1 Two Wheeler 500  
2 Four Wheeler 1000  
3 Three Wheeler 700

Design:

**Vehicle Service Station**

**SERVICE DETAILS**

Service ID

Vehicle number

Service Details

Vehicle Type

**VEHICLE TYPE**

Vehicle number

Vehicle Type

Service Charge

**Insert** **Update**

Total Amount

Vehicle Service Station

### SERVICE DETAILS

Service ID:

Vehicle number:

Service Details:

Vehicle Type:

### VEHICLE TYPE

Vehicle number:

Vehicle Type:

Service Charge:

Total Amount:

#### Code:

```
using System.Data;
using System.Data.SqlClient;

namespace VehicleServiceStation
{
    public partial class Form1 : Form
    {
        SqlConnection con = new SqlConnection(@"Data Source=NITHIN-LAP\SQLEXPRESS;Initial Catalog = master; Integrated Security = True");
        SqlCommand cmd = new SqlCommand();
        SqlDataReader dr;
        SqlDataAdapter da;
        DataSet ds = new DataSet();
        DataTable dt = new DataTable();

        public Form1()
        {
            InitializeComponent();
        }

        private void Form1_Load(object sender, EventArgs e)
        {
            cmd.Connection = con;
            loadlist();
        }

        public void loadlist()
        {
            con.Open();
            cmd.CommandText = "select * from tbl_VehicleType";
            dr = cmd.ExecuteReader();
            if (dr.HasRows)
            {
                while (dr.Read())
            }
        }
    }
}
```



```

        {
            comboBox1.Items.Add(dr[1].ToString());
        }
    }
    con.Close();
    con.Open();
    cmd.CommandText = "select * from tbl_ServiceDetails";
    dr = cmd.ExecuteReader();
    if (dr.HasRows)
    {
        while (dr.Read())
        {
            comboBox2.Items.Add(dr[1].ToString());
        }
    }
    con.Close();
}

private void button1_Click(object sender, EventArgs e)
{
    con.Open();
    cmd.CommandText = "insert into tbl_ServiceDetails values('" + textBox1.Text + "','" +
    textBox2.Text + "','" + textBox3.Text + "','" + comboBox1.SelectedItem.ToString() + "')";
    cmd.ExecuteNonQuery();
    cmd.Clone();
    MessageBox.Show("Record inserted");
    con.Close();
    textBox1.Text = "";
    textBox2.Text = "";
    textBox3.Text = "";
}
String a;
private void button2_Click(object sender, EventArgs e)
{
    con.Open();
    cmd.CommandText = "select * from tbl_VehicleType where VehicleType='" +
    textBox4.Text + "'";
    dr = cmd.ExecuteReader();
    if (dr.HasRows)
    {
        while (dr.Read())
        {
            {
                a = dr[2].ToString();
            }
        }
    }
    con.Close();
    textBox6.Text = Convert.ToString(Convert.ToInt64(textBox5.Text) +
    Convert.ToInt64(a));
}

private void comboBox2_SelectedIndexChanged(object sender, EventArgs e)
{
    con.Open();
    cmd.CommandText = "select * from tbl_ServiceDetails where VehicleNumber='" +

```

```
comboBox2.SelectedItem.ToString() + """,  
dr = cmd.ExecuteReader();  
if (dr.HasRows)  
{  
    while (dr.Read())  
    {  
        textBox4.Text = dr[3].ToString();  
    }  
}  
con.Close();  
}  
}
```